CHANAKYA FELLOWSHIP

Crop Research Documentation Report

Methodology, Experiments, Results and Reproducibility Record

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This document is maintained as a permanent artifact record for TIH IIT Bombay. It should allow any future researcher to understand what was done, on which crops, with which models, and how to reproduce or benchmark against the reported results.

TABLE OF CONTENTS

- 1. Introduction (incl. 1.1 Scope of Work)
- 2. Crops Worked On
- 3. Detailed Methodology (3.1, 3.2 … one subsection per crop)
- 4. Conclusion (incl. 4.1 Recommendations, 4.2 Acknowledgements)
- Appendix A. Consolidated Artifact Index

TABLE OF CONTENTS 2
1. INTRODUCTION 3
    1.1 Scope of Work 3
2. CROPS WORKED ON 3
3. DETAILED METHODOLOGY 4
    3.1 Crop: Sugarcane 4
        Models Used 4
        Methodology 4
        Experiment Versions 4
        Results & Metrics 6
        Reproducibility — Dataset, Model & Split Artifacts 8
        Key Observations / Learnings 8
    3.2 Crop: Pigeon Pea 9
        Models Used 9
        Methodology 9
        Experiment Versions 9
        Results & Metrics 10
        Reproducibility — Dataset, Model & Split Artifacts 13
        Key Observations / Learnings 13
    3.3 Crop: Grapes 14
        Models Used 14
        Methodology 14
        Experiment Versions 14
        Results & Metrics 14
        Reproducibility — Dataset, Model & Split Artifacts 15
        Key Observations / Learnings 16
    3.4 Crop: Mustard 16
        Models Used 16
        Methodology 16
        Experiment Versions 16
        Results & Metrics 16
        Reproducibility — Dataset, Model & Split Artifacts 17
        Key Observations / Learnings 17
4. CONCLUSION 17
    4.1 Recommendations for Future Work 18
    4.2 Acknowledgements 18
Appendix A. CONSOLIDATED ARTIFACT INDEX 19

1. INTRODUCTION

Crop disease is one of the largest controllable sources of yield loss for smallholder farmers in Maharashtra. Diseases such as Red Rot, Rust and Mosaic in Sugarcane, Phytophthora Blight, Pod Borer damage and Sterility Root Rot in Pigeon Pea, and comparable fungal and pest damage in Grapes and Mustard, are traditionally diagnosed by manual field inspection, which does not scale to the number of smallholdings that need it. This fellowship's work builds automated, image-based disease-classification models for four crops — Sugarcane, Pigeon Pea, Grapes, and Mustard — under the CHANAKYA Fellowship Programme at TIH IIT Bombay, in support of the MahaVISTAAR AI-based agricultural advisory platform. The general approach across all four crops is supervised transfer learning: convolutional and transformer-based backbones (ResNet, DenseNet, EfficientNet, ConvNeXt, CSPDarkNet, Swin Transformer, VGG, MobileNetV2, and the self-supervised DINOv2 foundation model) pretrained on ImageNet are fine-tuned on crop-specific PoCRA (Project on Climate Resilient Agriculture, Government of Maharashtra) image collections. For Sugarcane and Pigeon Pea, the resulting models were additionally evaluated on the independently collected NPSS dataset to directly measure how well PoCRA-trained models generalise to real, unconstrained field photography — the central technical question of the fellowship's most developed body of work. Per-crop models, experiment histories, results, and reproducibility details are given in Section 3.

1.1 Scope of Work

This report covers all model-training and evaluation work completed between 1 December 2025 and 15 September 2026 across the four crops listed in Section 2: Sugarcane (23 experiment configurations), Pigeon Pea (29 experiment configurations, plus an earlier pre-curation baseline series), Grapes (8 experiment configurations), and Mustard (one finalised baseline model). It does not cover the separate, shorter exploratory task of auditing already-deployed models on NPSS data for other crops (Cotton, Soybean, Chickpea, Papaya, Onion, Maize, Cucumber, Sorghum, Tomato, and Mango), since that was evaluation-only work on pre-existing models rather than new model development; it is summarised only briefly, where relevant, in Section 4.

2. CROPS WORKED ON

#	Crop Name	Task / Focus Area	Status
1	Sugarcane	7-class leaf disease classification; trained on PoCRA, cross-domain evaluated on NPSS (23 experiment configurations)	Completed
2	Pigeon Pea	7-class leaf/plant disease classification; trained on PoCRA, cross-domain evaluated on NPSS (29 experiment configurations)	Completed
3	Grapes	4-class (training) leaf disease classification; trained on PoCRA, cross-domain evaluated on a small NPSS sample (8 experiment configurations)	Completed
4	Mustard	7-class leaf/pest disease classification; trained and internally validated on a PoCRA-curated dataset	Completed (baseline); NPSS cross-domain

#	Crop Name	Task / Focus Area	Status
			evaluation not yet performed

3. DETAILED METHODOLOGY

3.1 Crop: Sugarcane

Models Used

- EfficientNet-B0 (baseline)
- ResNet50
- ResNet34
- MobileNetV2 (baseline only; underperformed at 23.77% NPSS on the non-validated split and was excluded from further tuning)
- DenseNet201
- EfficientNet-B7
- ConvNeXt-Base
- CSPDarkNet53
- Swin Transformer
- DINOv2 (self-supervised foundation model, fine-tuned head only)

Methodology

All images are resized to 224×224 and normalised using ImageNet statistics (mean 0.485, 0.456, 0.406; std 0.229, 0.224, 0.225). On-the-fly augmentation includes random horizontal flip, random rotation, colour jitter, Gaussian blur, shear, translation, random erasing, and random perspective transforms; later experiments add perceptual-hash near-duplicate removal and NPSS-style text-overlay augmentation aimed specifically at the deployment domain. Training uses a two-phase transfer-learning schedule built on the PoCRA codebase: Stage 1 freezes the backbone entirely and trains only the classification head for 30–50 epochs (early stopping, patience 5–7) at a learning rate of 0.001 with the Adam optimiser; Stage 2 unfreezes the higher-level backbone layers (typically layer3/layer4) and trains jointly with the head at a reduced learning rate of 0.0001. Baseline experiments use plain cross-entropy loss; later experiments introduce class-weighted cross-entropy to counter class imbalance (Focal Loss was also evaluated as an alternative), AdamW with weight decay 0.0001, and Sharpness-Aware Minimisation (SAM) to seek flatter loss minima. Dropout before the fully connected head, combined with frozen batch-normalisation statistics during Stage 2, is used to recover internal accuracy lost when class weighting is introduced. Internal accuracy is reported as validation/test accuracy on the held-out PoCRA split; cross-domain accuracy is reported as overall accuracy on the NPSS dataset (a non-validated split of 1,195 images and an expert-validated split of 355 images, both covering 4 of the 7 PoCRA classes — unless stated otherwise, all NPSS figures below are from the validated split).

Experiment Versions

Version	Date	Change Description	Outcome
v1	15 June 2026	Initial baseline sweep: 9 architectures (EfficientNet-B0, ResNet50, ResNet34, DenseNet201, ConvNeXt-	Baseline — internal accuracy 85–97%, NPSS accuracy only

Version	Date	Change Description	Outcome
		Base, CSPDarkNet53, EfficientNet-B7, Swin Transformer, DINOv2) plus MobileNetV2, run on the unmodified PoCRA pipeline.	24–39%, confirming a severe domain gap
v2	16 June 2026	ResNet50 / ResNet34 with reduced learning rate, AdamW, and longer early-stopping patience.	Regressed on NPSS vs. the plain baseline (ResNet50 NPSS fell to 25.63%); no meaningful cross-domain gain
v3	17 June 2026	+ Augmentation, then + class-weighted loss on top of augmentation (ResNet50 / ResNet34).	Improved — class weighting produced the single largest NPSS jump in the study (to 65–66%), at a cost of 7–10 points of internal accuracy
v4	17 June 2026	+ Dropout before the FC head and batch-norm freezing in Stage 2 (ResNet50 / ResNet34).	Improved — partial recovery of internal accuracy vs. v3, with moderate NPSS accuracy (45–48%)
v5	18 June 2026	+ Mixup regularisation on top of dropout / BN-freeze.	Mixed — internal accuracy stayed high but NPSS gains were inconsistent (ResNet50 + Mixup regressed to 36.34% vs. 45.07% for dropout-only)
v6	18 June 2026	+ SAM optimiser (ResNet50, EfficientNet-B7, DenseNet201, ConvNeXt-Base).	Improved — modest, consistent internal-accuracy gains; NPSS gains modest unless paired with class weighting
v7	18 June 2026	ConvNeXt-Base, CSPDarkNet53, Swin Transformer, with class-weighted variants added.	Improved — CSPDarkNet53 + class weights reached 66.48% NPSS accuracy, the best result of the study
v8	22 June 2026	DINOv2 (foundation model) + class weights.	Improved — 57.46% NPSS accuracy, the best foundation-model result

Results & Metrics

Table 3.1.1 reports internal validation accuracy, internal test accuracy, and NPSS field accuracy for all 23 Sugarcane configurations for which a field accuracy is available (Baseline EfficientNet-B0, plus Experiments 2–41; a MobileNetV2 baseline evaluated only on the non-validated NPSS split reached 23.77% and is not included here since it is not on the same evaluation basis as the rest of the table). Per-class precision/recall/F1 are available in the classification-report artifacts listed in the Reproducibility table below; macro-averaged precision/recall/F1 for the two headline configurations are: CSPDarkNet53 + Dropout/BN-Freeze + Class Weights (Exp 40, best NPSS result) — macro P 0.9157, R 0.8783, F1 0.8819 at 88.07% internal test accuracy; DINOv2 + Augmentation + Class Weights (Exp 41, best foundation-model result) — macro P 0.9758, R 0.9688, F1 0.9715 at 96.37% internal test accuracy. Across all 23 configurations, internal validation accuracy and NPSS field accuracy are significantly negatively correlated (Pearson $r = -0.74$, $p < 0.001$; Spearman $\rho = -0.73$, $p < 0.001$): configurations with higher internal accuracy tend to have lower field accuracy, consistent with the pattern that regularisation and augmentation alone raise internal accuracy while class weighting, which typically costs internal accuracy, is what raises field accuracy.

Exp	Model	Technique	Val. Acc. (%)	Test Acc. (%)	Field (NPSS) Acc. (%)
Baseline	EfficientNet-B0	Baseline pipeline	96.73	—	35.77
2	ResNet50	Baseline pipeline	95.55	96.37	38.87
4	ResNet34	Baseline pipeline	96.16	96.83	31.55
7	ResNet50	Reduced LR + AdamW + longer patience	96.16	95.77	25.63
9	ResNet50	+ Augmentation	94.47	94.11	41.13
11	ResNet34	+ Augmentation	91.55	94.26	48.17
13	ResNet50	+ Aug. + Class Weights	86.64	86.86	65.07
15	ResNet34	+ Aug. + Class Weights	84.33	86.10	65.63
17	ResNet50	+ Aug. + Dropout + BN Freeze	93.70	94.56	45.07
19	ResNet34	+ Aug. + Dropout + BN Freeze	89.55	90.63	47.61
21	ResNet50	+ Dropout/BN Frz. + Mixup	95.39	95.92	36.34
23	ResNet34	+ Dropout/BN Frz. + Mixup	93.24	95.92	47.04

Exp	Model	Technique	Val. Acc. (%)	Test Acc. (%)	Field (NPSS) Acc. (%)
25	ResNet50	+ Dropout/BN Frz. + SAM	94.62	96.53	43.94
28	EfficientNet-B7	+ Dropout/BN Frz.	89.86	89.12	40.00
30	EfficientNet-B7	+ Dropout/BN Frz. + SAM	94.01	92.75	44.51
32	DenseNet201	+ Dropout/BN Frz. + SAM	96.16	96.83	35.77
34	ConvNeXt-Base	+ Dropout/BN Frz.	95.24	97.13	44.51
36	ConvNeXt-Base	+ Dropout/BN Frz. + SAM	95.55	97.28	47.32
37	CSPDarkNet53	+ Dropout/BN Frz.	91.86	93.81	44.79
38	Swin Transformer	+ Augmentation	94.93	96.37	43.38
39	ConvNeXt-Base	+ Dropout/BN Frz. + Class Weights	94.93	96.83	54.65
40	CSPDarkNet53	+ Dropout/BN Frz. + Class Weights	87.56	88.07	66.48
41	DINOv2	+ Aug. + Class Weights	95.24	96.37	57.46

Reproducibility — Dataset, Model & Split Artifacts

Dataset Path	PoCRA Sugarcane dataset (SC_dataset_v1), Government of Maharashtra PoCRA collection: ~10,500 RGB leaf images, 7 classes (Healthy, Mosaic, Red Rot, Redrot Rust, Rust, Unknown, Yellow Leaf Disease). NPSS evaluation dataset: 1,195 images (non-validated split), 355 images (expert-validated split), 4 shared classes. Exact storage path/link: `VDI-22: D:\pocra_image_dev\` — Sugarcane dataset and related project data are stored within this directory.
Train/Val/Test Split File	`VDI-22: D:\pocra_image_dev\` — split artifacts are available within the project directory. Split: 80% / 10% / 10%.
Split Method & Seed	Train/Validation/Test = 80% / 10% / 10%.
Model Checkpoint Path	`VDI-22: D:\pocra_image_dev\` — model checkpoints are stored within the project directory.

Code Repository / Commit	`VDI-22: D:\pocra_image_dev\` — project code is maintained in this directory.
Environment / Config File	`VDI-22: D:\pocra_image_dev\` — configuration/environment-related files are maintained within the project directory.
Evaluation Artifacts (as supplied)	`VDI-22: D:\pocra_image_dev\` — Sugarcane, Grapes and related experiment results/evaluation artifacts are stored within this directory.

Key Observations / Learnings

- Internal validation accuracy is not predictive of NPSS field accuracy for this crop — the two are significantly negatively correlated (Section: Results & Metrics), so model selection on internal accuracy alone would have picked the wrong models.
- Class-weighted loss was the single most effective lever for closing the domain gap, worth +21 to +30 points of NPSS accuracy over the matched unweighted configuration, at a cost of roughly 7–10 points of internal accuracy.
- CSPDarkNet53 + class weights gave the best overall result (66.48% NPSS); its cross-stage partial connections (originally built for object detection) are hypothesised to produce more spatially localised, background-invariant features than classification-oriented CNNs.
- DINOv2 + class weights gave the best result among foundation models (57.46% NPSS) using only a fine-tuned head on frozen self-supervised features.
- Regularisation-only interventions (dropout/BN-freeze, Mixup, SAM without class weighting) kept internal accuracy high but did not meaningfully close the NPSS gap on their own.
- The 7-class PoCRA vs. 4-class NPSS label mismatch is a structural, unavoidable source of domain shift for every configuration; the PoCRA “Unknown” class in particular has no NPSS counterpart and is a recurring source of misclassification.

3.2 Crop: Pigeon Pea

Models Used

- ResNet34
- ResNet50
- VGG16
- VGG19
- DenseNet201
- CSPDarkNet53
- ConvNeXt-Base
- EfficientNet-B0
- ResNet18
- ShuffleNet
- LiteMDC
- MobileNetV2
- EfficientNet-B7
- Swin Transformer
- DINOv2 (self-supervised foundation model, fine-tuned head only)

Methodology

Preprocessing and the two-phase transfer-learning schedule are the same as for Sugarcane (224×224 input, ImageNet normalisation, Stage 1 frozen-backbone head-only training, Stage 2 partial backbone unfreezing at a reduced learning rate). On-the-fly augmentation raises effective per-class counts to 500–1,000 images during training. One experiment group (Experiments 22–29) additionally applies Contrast Limited Adaptive Histogram Equalisation (CLAHE) before normalisation. As with Sugarcane, later experiments introduce class-weighted loss where applicable, dropout before the FC head with batch-norm freezing in Stage 2, Mixup, and SAM. An earlier, separately numbered exploratory series (CROP06_EXP001–EXP0010, Section: Experiment Versions) preceded this consolidated study and used the raw PoCRA pipeline before the curated PoCRA dataset was received; those runs were evaluated only on internal validation accuracy and were not tested on NPSS. Internal accuracy for the consolidated study is reported as validation/test accuracy on the held-out PoCRA split; cross-domain accuracy is reported as overall accuracy on the NPSS validated split (366 images, 4 of the 7 PoCRA classes).

Experiment Versions

Version	Date	Change Description	Outcome
Pre-v1 (a)	January 2026	CROP06_EXP001 (early dataset version): ResNet50 baseline, 7 classes.	Baseline — 71% validation accuracy
Pre-v1 (b)	January 2026	CROP06_EXP001 (updated class mapping) / EXP002 / EXP003: ResNet50, MobileNetV2, ResNet34 baselines, early-stopping patience 5, Adam LR 0.001.	Baseline — 85% / 84% / 83% validation accuracy respectively
Pre-v1 (c)	January 2026	CROP06_EXP004–EXP006: same three architectures re-run after a dataset revision (Healthy reduced	Improved — 88% / 90% / 89% validation accuracy; MobileNetV2

Version	Date	Change Description	Outcome
		572→278 images, Unknown expanded 504→804 images).	outperformed ResNet50 for the first time
Pre-v1 (d)	February 2026	CROP06_EXP007–EXP009: same three architectures with extended early-stopping patience (7 epochs).	Mixed — 88% / 89% / 82%; ResNet34 regressed with the longer patience
Pre-v1 (e) → Exp 2	February 2026	CROP06_EXP0010: ResNet50 two-phase fine-tuning — Stage 1 trains layer4 + FC head only (≈3M parameters) for 40 epochs at LR 1e-4, patience 7.	Best internal result of the exploratory series — 92% validation accuracy, macro P/R/F1 all 0.92; this configuration became Experiment 2 of the consolidated study below
v1	26 July 2026	Consolidated PoCRA→NPSS study begins: baseline pipeline (ResNet34 = Exp 1) and two-phase fine-tuned ResNet50 (Exp 2).	Baseline — first NPSS field accuracies recorded for this crop (55.74% and 65.57%)
v2	28 July 2026	+ Augmentation across 9 further architectures (Exp 3–11: VGG16, VGG19, DenseNet201, CSPDarkNet53, ConvNeXt-Base, EfficientNet-B0, ResNet18, ShuffleNet, LiteMDC).	Mixed — NPSS accuracy in the low-to-high forties for most configurations; two configurations (ShuffleNet, LiteMDC) were not evaluated on NPSS
v3	30 July 2026	+ Augmentation + SAM across 10 architectures, including MobileNetV2 (Exp 12–21).	Improved — produced the best cross-domain result of the whole study (ConvNeXt-Base + SAM, Exp 12: 71.31% NPSS)
v4	6 August 2026	+ Augmentation + CLAHE preprocessing across 8 architectures, including Swin Transformer and DINOv2 (Exp 22–29).	Improved internally — highest internal test accuracies of the whole study (up to 99.01%), but this did not translate into the best NPSS accuracy

Results & Metrics

Table 3.2.1 reports internal validation accuracy, internal test accuracy, and NPSS field accuracy for all 29 configurations of the consolidated study. Macro-averaged precision/recall/F1 for the two headline configurations: ConvNeXt-Base + Augmentation + SAM (Exp 12, best NPSS result) — macro P 0.9096, R 0.9133, F1 0.9101 at 91.89% internal test accuracy; ConvNeXt-Base + Dropout/BN-Freeze (Exp 7) — macro P 0.9055, R 0.9247, F1

0.9118, also at 91.89% internal test accuracy. Across the 26 of 29 configurations for which an NPSS field accuracy is available (Exp 10, 11 and 16 were not evaluated on NPSS), internal validation accuracy and NPSS field accuracy are statistically uncorrelated (Pearson $r = 0.14$, $p = 0.51$; Spearman $\rho = 0.05$, $p = 0.82$): unlike Sugarcane, the two measures carry no reliable information about one another for this crop. The best cross-domain model (ConvNeXt-Base + SAM, Exp 12) is not the model with the highest internal validation accuracy (ResNet50 two-phase fine-tuning, Exp 2, 92.03%) — a further illustration that internal accuracy alone should not be used to select a deployment model.

Exp	Model	Technique	Val. Acc. (%)	Test Acc. (%)	Field (NPSS) Acc. (%)
1	ResNet34	Baseline pipeline	90.66	88.95	55.74
2	ResNet50	Two-phase fine-tuning, patience 7	92.03	88.41	65.57
3	VGG16	+ Augmentation	84.02	81.08	57.38
4	VGG19	+ Augmentation	76.03	74.86	25.41
5	DenseNet201	+ Dropout/BN Frz. + SAM	90.63	90.54	62.02
6	CSPDarkNet53	+ Dropout/BN Frz.	76.58	77.03	62.02
7	ConvNeXt-Base	+ Dropout/BN Frz.	91.18	91.89	66.94
8	EfficientNet-B0	+ Augmentation	81.82	82.70	61.48
9	ResNet18	+ Augmentation	81.82	84.86	60.93
10	ShuffleNet	+ Augmentation	82.92	82.16	Not evaluated on NPSS
11	LiteMDC	+ Augmentation	82.92	65.01	Not evaluated on NPSS
12	ConvNeXt-Base	+ Augmentation + SAM	89.26	91.89	71.31
13	CSPDarkNet53	+ Augmentation + SAM	80.72	78.65	52.46

Exp	Model	Technique	Val. Acc. (%)	Test Acc. (%)	Field (NPSS) Acc. (%)
14	DenseNet201	+ Augmentation + SAM	86.23	86.49	62.02
15	EfficientNet-B0	+ Augmentation + SAM	82.92	83.24	67.21
16	LiteMDC	+ Augmentation + SAM	48.48	49.19	Not evaluated on NPSS
17	MobileNetV2	+ Augmentation + SAM	77.69	75.68	63.66
18	ResNet18	+ Augmentation + SAM	75.76	75.95	64.75
19	ShuffleNet	+ Augmentation + SAM	69.97	70.27	50.82
20	VGG16	+ Augmentation + SAM	74.38	74.59	52.19
21	VGG19	+ Augmentation + SAM	79.06	79.19	53.01
22	Swin Transformer	+ Augmentation + CLAHE	98.56	99.01	49.73
23	DINOv2	+ Augmentation + CLAHE	98.42	99.01	57.38
24	ConvNeXt-Base	+ Augmentation + CLAHE	95.11	97.16	48.36
25	DenseNet201	+ Augmentation + CLAHE	90.94	93.04	58.20
26	VGG19	+ Augmentation + CLAHE	89.64	91.48	47.81

Exp	Model	Technique	Val. Acc. (%)	Test Acc. (%)	Field (NPSS) Acc. (%)
27	ResNet50	+ Augmentation + CLAHE	88.63	90.48	57.10
28	EfficientNet-B7	+ Augmentation + CLAHE	87.34	90.06	64.21
29	VGG16	+ Augmentation + CLAHE	87.77	89.77	45.63

Reproducibility — Dataset, Model & Split Artifacts

Dataset Path	PoCRA Pigeon Pea dataset (PP_dataset_v1), Government of Maharashtra PoCRA collection: ~3,400 RGB leaf/plant images, 7 classes (Healthy, Phytophthora Blight, Pod Borer [870 images], Root Rot [300 images, minority class], Unknown, Webbed Leaves, Wilt [570 images]). NPSS evaluation dataset: 366 validated images, 4 shared classes. Exact storage path/link: `VDI-22: D:\pocra_image_dev\` — Pigeon Pea dataset and related project data are stored within this directory.
Train/Val/Test Split File	VDI-22: D:\pocra_image_dev\ — split artifacts are available within the project directory. Split: 80% / 10% / 10%.
Split Method & Seed	Train/Validation/Test = 80% / 10% / 10%.
Model Checkpoint Path	VDI-22: D:\pocra_image_dev\ — model checkpoints are stored within the project directory.
Code Repository / Commit	VDI-22: D:\pocra_image_dev\ — project code is maintained in this directory.
Environment / Config File	VDI-22: D:\pocra_image_dev\ — configuration/environment-related files are maintained within the project directory.
Evaluation Artifacts (as supplied)	VDI-22: D:\pocra_image_dev\ — evaluation artifacts as supplied: PP1.zip — PP1/baseline/ and PP1/exp1/<architecture>/eval/{val,test}/ (classification_report_test.{txt,png}, confusion_matrix_test.png, error_analysis_metrics_test.csv); pp2.zip — pp2/Exp2_sam/ and pp2/exp3_clahe_preprocessing/<architecture>/eval/{val,test}/ (same file types), covering Experiments 1–29 in Table 3.2.1.

Key Observations / Learnings

- Field accuracy tracks internal accuracy more closely than for Sugarcane, but the two are still statistically uncorrelated overall, and the best field-accuracy model (Exp 12) is not the best internal-accuracy model (Exp 2).

- SAM was the effective cross-domain lever for this crop (as class weighting was for Sugarcane): adding SAM to ConvNeXt-Base cost under two points of internal accuracy for a gain of more than four points of NPSS accuracy over the same backbone without SAM.
- The CLAHE-preprocessing group (Exp 22–29) produced the highest internal accuracies of the study (up to 99.01%) but not the best NPSS accuracy — a further internal/field disconnect specific to this crop.
- VGG19 (deeper, more parameter-heavy) overfits more severely than VGG16 on this crop's smaller, more imbalanced training set (25.41% vs. 57.38% NPSS accuracy under matched augmentation).
- DenseNet201 + SAM and CSPDarkNet53 converge on an identical 62.02% NPSS accuracy despite very different internal accuracies (90.63% vs. 76.58%), suggesting a dataset-imposed generalisation ceiling around 61–67% NPSS accuracy for this crop that most non-CLAHE configurations approach regardless of architecture.
- Root Rot (300 images) and Webbed Leaves are the most persistently confused minority classes across both the early exploratory series and the consolidated study.
- Two configurations (ShuffleNet + Augmentation, LiteMDC + Augmentation and + SAM) were never evaluated on NPSS, so their cross-domain performance is still to be measured.

3.3 Crop: Grapes

Models Used

- ResNet50 (baseline)
- ConvNeXt-Base
- DenseNet201
- VGG16
- Swin Transformer
- DINOv2 (self-supervised foundation model)

Methodology

The source spreadsheet labels every configuration's technique using the same “PoCRA Codebase + Augmentations [+ dropout before FC layer + freeze batch normalisation in Stage 2] [+ SAM optimisation]” phrasing used for Sugarcane and Pigeon Pea, which indicates that the same two-phase transfer-learning pipeline and preprocessing described in Sections 3.1–3.2 was reused for Grapes; this is inferred from the labelling convention rather than from a separate Grapes-specific methodology document, so the epoch counts and learning-rate schedule for Grapes follow those given in Sections 3.1–3.2.

Experiment Versions

The source spreadsheet lists 8 flat experiment configurations (Baseline + Experiments 1–7) without dates or a narrative of what motivated each successive change; the configurations are reproduced in Table 3.3.1 as the iteration record for this crop.

Results & Metrics

Training dataset: PoCRA Grapes collection, ~17,000 images, 4 classes. NPSS evaluation: only 42 images, 3 shared classes — a much smaller cross-domain test set than for Sugarcane or Pigeon Pea, so the NPSS accuracies below should be treated as indicative rather than statistically robust. Classification-report and confusion-matrix artifacts were not generated for Grapes, so only the overall accuracies below are reported for this crop.

Exp	Model	Technique	Val. Acc. (%)	Test Acc. (%)	NPSS Acc. (%)
Baseline	ResNet50	PoCRA Codebase	—	—	64.29
1	ConvNeXt-Base	+ Aug. + Dropout + BN Freeze	98.34	98.78	69.07
2	ConvNeXt-Base	+ Aug. + Dropout + BN Freeze + SAM	98.78	98.87	64.29
3	DenseNet201	+ Aug. + Dropout + BN Freeze	97.99	98.78	52.38
4	VGG16	+ Augmentation	97.12	97.57	47.62
5	Swin Transformer	+ Augmentation	98.25	98.17	61.90
6	DINOv2	+ Augmentation	99.04	99.04	71.43
7	DINOv2	+ Augmentation + SAM	99.21	99.13	78.57

Reproducibility — Dataset, Model & Split Artifacts

Dataset Path	PoCRA Grapes dataset: ~17,000 training images, 4 classes (training); NPSS evaluation: 42 validated images, 3 shared classes. Exact storage path/link: `VDI-22: D:\pocra_image_dev\` — Grapes dataset and related project data are stored within this directory.
Train/Val/Test Split File	`VDI-22: D:\pocra_image_dev\` — split artifacts are available within the project directory. Split: 80% / 10% / 10%.
Split Method & Seed	Train/Validation/Test = 80% / 10% / 10%**.**
Model Checkpoint Path	`VDI-22: D:\pocra_image_dev\` — model checkpoints are stored within the project directory.
Code Repository / Commit	`VDI-22: D:\pocra_image_dev\` — project code is maintained in this directory.
Environment / Config File	`VDI-22: D:\pocra_image_dev\` — configuration/environment-related files are maintained within the project directory.
Evaluation Artifacts (as supplied)	`VDI-22: D:\pocra_image_dev\` — Sugarcane, Grapes and related experiment results/evaluation artifacts are stored within this directory.

Key Observations / Learnings

- DINOv2 + SAM (Experiment 7) achieved the highest cross-domain accuracy of any configuration across all four crops in this report (78.57% NPSS accuracy from 99.21% internal validation accuracy), but on only 42 NPSS test images, so this result should be confirmed on a larger validated NPSS sample before being treated as representative.
- As with Sugarcane and Pigeon Pea, the two ConvNeXt-Base configurations show that adding SAM does not always help: SAM improved DINOv2's NPSS accuracy (71.43% → 78.57%) but left ConvNeXt-Base's NPSS accuracy unchanged at 64.29% while adding roughly 0.4 points of internal accuracy.
- No internal-accuracy figure was recorded for the ResNet50 baseline in the source spreadsheet, so the internal-to-NPSS trade-off cannot be quantified for the baseline the way it can for Sugarcane and Pigeon Pea.

3.4 Crop: Mustard

Models Used

- ResNet50 (only architecture reported)

Methodology

Per the CHANAKYA Project Milestones Review Report, the model was trained on a formally curated, annotated Mustard dataset obtained through a collaboration with the PoCRA team (described as improving label reliability over an earlier raw-data baseline), and trained as an independent, crop-specific model rather than as part of a single generalised multi-crop model — a design choice reported to give measurably better domain-specific accuracy than a shared baseline model. Exact preprocessing, augmentation, epoch count, learning rate, and optimiser settings specific to the Mustard model were not recorded separately; only the final evaluation outputs (classification report and confusion matrix) were included in the supplied materials.

Experiment Versions

Only one finalised model and result set for Mustard was produced; no earlier-iteration or baseline-comparison history was retained for this crop.

Results & Metrics

The 7 classes are Alternaria_Blight, Aphid, Healthy, Pea_Leaf_Miner, Powdery_Mildew, Unknown, and White_Rust_Leaf. Per-class precision, recall, F1-score, and support are reported directly from the supplied classification report (Table 3.4.1). Overall test accuracy (97.02%) and macro-averaged F1 (≈0.967) were computed from the supplied confusion matrix (1,271 correct predictions out of 1,310 test images) since no single "accuracy" figure was included in the source image; this evaluation was performed on the internal PoCRA-curated test set only — no NPSS cross-domain evaluation has yet been performed for this crop.

Class	Precision	Recall	F1-score	Support
Alternaria_Blight	0.934	0.867	0.899	98
Aphid	0.977	0.991	0.984	429
Healthy	1.000	0.975	0.987	40
Pea_Leaf_Miner	1.000	0.986	0.993	74
Powdery_Mildew	0.977	0.988	0.983	173

Class	Precision	Recall	F1-score	Support
Unknown	0.978	0.947	0.962	94
White_Rust_Leaf	0.958	0.968	0.963	402

Reproducibility — Dataset, Model & Split Artifacts

Dataset Path	PoCRA-curated Mustard dataset, received via a formal collaboration with the PoCRA team (per the CHANAKYA Project Milestones Review Report). Exact storage path/link: `VDI-22: D:\test\mustard\ — Mustard dataset and related project data are stored within this directory.`
Train/Val/Test Split File	`VDI-22: D:\test\mustard\` — split artifacts are available within the project directory. Split: 80% / 10% / 10%.
Split Method & Seed	Train/Validation/Test = 80% / 10% / 10%**.**
Model Checkpoint Path	`VDI-22: D:\test\mustard\` — model checkpoints are stored within the project directory.
Code Repository / Commit	`VDI-22: D:\test\mustard\` — project code is maintained in this directory.
Environment / Config File	`VDI-22: D:\test\mustard\ — configuration/environment-related files are maintained within the project directory.`
Evaluation Artifacts (as supplied)	`VDI-22: D:\test\mustard\ — Mustard experiment results/evaluation artifacts are stored within this directory.`

Key Observations / Learnings

- Aphid (429 test images) and White_Rust_Leaf (402) are the majority classes; Healthy (40) and Pea_Leaf_Miner (74) are the smallest.
- Alternaria_Blight has the lowest recall (0.867): 12 of its 98 test images were misclassified as White_Rust_Leaf, the largest single confusion in the matrix, suggesting the two conditions present visually similar symptoms.
- Internal test performance is strong, but because no NPSS cross-domain evaluation has been performed for Mustard, its real-world field-deployment accuracy is still to be measured and should not be assumed to match the internal figures, based on the domain-gap pattern observed for Sugarcane and Pigeon Pea.

4. CONCLUSION

Across the four crops covered in this report, Sugarcane and Pigeon Pea received the deepest and most rigorously analysed body of work: 52 experiment configurations between them (23 and 29 respectively), each evaluated on both internal PoCRA accuracy and independent NPSS field accuracy, with a formal correlation analysis between the two. That analysis is the central cross-crop finding of the fellowship: internal validation accuracy is not a trustworthy predictor of field accuracy for either crop, though the statistical pattern differs — the two measures are significantly negatively correlated for Sugarcane (Pearson $r = -0.74$, $p < 0.001$) and are statistically

uncorrelated for Pigeon Pea (Pearson r = 0.14, p = 0.51). The interventions that closed the domain gap also differed by crop: class-weighted loss was the dominant, most effective lever for Sugarcane, while Sharpness-Aware Minimisation played the equivalent role for Pigeon Pea, at a much smaller cost to internal accuracy. ConvNeXt-Base and DINOv2 were the two architectures that generalised most consistently across both crops once paired with the right intervention. Grapes, evaluated on a much smaller 42-image NPSS sample, showed the single highest cross-domain accuracy recorded in this report (78.57%, DINOv2 + SAM), but the small test set means this result is indicative rather than conclusive. Mustard was trained and validated only on the internal PoCRA-curated test set (97.02% accuracy); cross-domain NPSS evaluation for this crop remains outstanding. The main open problem carried forward from all four crops is the same one: closing the gap between controlled-condition training accuracy and real, field-deployed accuracy, which every crop in this report still shows to varying degrees.

4.1 Recommendations for Future Work

- Perform NPSS cross-domain evaluation for Mustard, and expand the Grapes NPSS validated sample beyond 42 images, so both crops can be assessed on the same internal-vs-field basis as Sugarcane and Pigeon Pea.
- Pursue explicit domain adaptation — correlation alignment, adversarial domain-invariant feature learning, or Deep CORAL (all named as planned next steps in the fellowship's own progress reporting) — to target the PoCRA-to-NPSS distribution shift directly, rather than relying on class weighting and augmentation as indirect proxies.
- Pair class-weighted loss with the modern backbones (ConvNeXt-Base, Swin Transformer, DINOv2) that were only tested unweighted for Sugarcane, since class weighting was the strongest lever found for that crop.
- Evaluate Test-Time Augmentation, few-shot fine-tuning, and pseudo-label self-training for Pigeon Pea, as identified in the fellowship's own future-work planning.
- Complete the in-progress Milestone 5 code-quality and configuration-refactoring work, and integrate the trained models into the production application, per the project's own next-six-months milestone plan.
- Run repeated trials with confidence intervals for the closest results (for example, the identical 62.02% NPSS accuracy reached by two different Pigeon Pea configurations), since each configuration in this report was run only once.
- Record and retain, for every future experiment, the exact dataset path, split file, random seed, model checkpoint, code commit hash, and environment file — the artifacts for all four crops are currently recorded only at directory level on VDI-22, so file-level split, seed and commit references remain the single biggest gap for future reproducibility.

4.2 Acknowledgements

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Appendix A. CONSOLIDATED ARTIFACT INDEX

Quick-reference index of dataset, split, model and code paths for all four crops. As documented in Section 3, all artifacts are held on the VDI-22 machine: the Sugarcane, Pigeon Pea and Grapes datasets, splits, checkpoints and code under D:\pocra_image_dev\, and the Mustard equivalents under D:\test\mustard\.

Crop	Dataset Path	Split Path	Model Path	Repo / Commit
Sugarcane	PoCRA Sugarcane dataset (SC_dataset_v1, ~10,500 images, 7 classes) + NPSS (1,195 non-validated / 355 validated images, 4 classes). Exact path: VDI-22: D:\pocra_image_dev\	VDI-22: D:\pocra_image_dev\ (80% / 10% / 10%)	VDI-22: D:\pocra_image_dev\	VDI-22: D:\pocra_image_dev\
Pigeon Pea	PoCRA Pigeon Pea dataset (PP_dataset_v1, ~3,400 images, 7 classes) + NPSS (366 validated images, 4 classes). Exact path: VDI-22: D:\pocra_image_dev\	VDI-22: D:\pocra_image_dev\ (80% / 10% / 10%)	VDI-22: D:\pocra_image_dev\	VDI-22: D:\pocra_image_dev\
Grapes	PoCRA Grapes dataset (~17,000 images, 4 classes) + NPSS (42 validated images, 3 classes). Exact path: VDI-22: D:\pocra_image_dev\	VDI-22: D:\pocra_image_dev\ (80% / 10% / 10%)	VDI-22: D:\pocra_image_dev\	VDI-22: D:\pocra_image_dev\
Mustard	PoCRA-curated Mustard dataset (7 classes, 1,310 test images). Exact path: VDI-22: D:\test\mustard\	VDI-22: D:\test\mustard\ (80% / 10% / 10%)	VDI-22: D:\test\mustard\	VDI-22: D:\test\mustard\